

Lecture 17.1: Chomsky Normal Form

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To convert the Context Free Grammar to Chomsky Normal Form, following concepts are needed:

1. Eliminate the variables that derive nothing:

Discover all variables that derive terminal strings. For all other variables, remove all productions in which they appear either on the left or the right.

For example,

Consider: $S \rightarrow AB, A \rightarrow aA \mid a, B \rightarrow AB$

Although A derives all strings of a 's, B derives no terminal strings.

Thus, S derives nothing, and the language is empty.

2. Eliminate unreachable variables:

Eliminate the variables that do not appear in any derivation from the start symbol.

Basis: We can reach S (the start symbol).

Induction: If we can reach A , and there is a production $A \rightarrow \alpha$, then we can reach all symbols of α .

Remove all other symbols which are unreachable from the Start state.

3. Eliminate all useless symbols by:

- Eliminate symbols that derive no terminal string.
- Eliminate unreachable symbols.

4. Eliminate ϵ -Productions:

We can almost avoid using productions of the form $A \rightarrow \epsilon$ (called ϵ -productions).

The problem is that ϵ cannot be in the language of any grammar that has no ϵ -productions.

Theorem: If L is a CFL, then $L - \{\epsilon\}$ has a CFG with no ϵ -productions.

Nullable Variables: To eliminate ϵ -productions, we first need to discover the **nullable variables** (eg. A) which are in the form $A \Rightarrow^* \epsilon$.

Basis: If there is a production $A \rightarrow \epsilon$, then A is nullable.

Induction: If there is a production $A \rightarrow \alpha$, and all symbols of α are nullable, then A is nullable.

Example:

$S \rightarrow AB, A \rightarrow aA \mid \epsilon, B \rightarrow bB \mid A$

Basis: A is nullable because of $A \rightarrow \epsilon$.

Induction: B is nullable because of $B \rightarrow A$.

Then, S is nullable because of $S \rightarrow AB$.

Key idea: turn each production $A \rightarrow X_1 \dots X_n$ into a family of productions.

For each subset of nullable X's, there is one production with those eliminated from the right side "in advance."

Except, if all X's are nullable, do not make a production with ϵ as the right side.

Example:

$S \rightarrow ABC, A \rightarrow aA \mid \epsilon, B \rightarrow bB \mid \epsilon, C \rightarrow \epsilon$

A, B, C, and S are all nullable.

New grammar:

$S \rightarrow AB \mid A \mid B$

$A \rightarrow aA \mid A$

$B \rightarrow bB \mid B$

5. Eliminate Unit Productions:

A **unit production** is one whose right side consists of exactly one variable.

These productions can be eliminated.

Key idea: If $A \Rightarrow^* B$ by a series of unit productions, and $B \rightarrow \alpha$ is a non-unit-production, then add production $A \rightarrow \alpha$.

Then, drop all unit productions.

6. Cleaning up the Grammar:

Start with a CFG for L.

Perform the following steps in order:

- Eliminate ϵ -productions.
- Eliminate unit productions.
- Eliminate variables that derive no terminal string.
- Eliminate variables not reached from the start symbol.

Chomsky Normal Form:

A CFG is said to be in *Chomsky Normal Form* if every production is of one of these two forms:

1. $A \rightarrow BC$ (right side is two variables).
2. $A \rightarrow a$ (right side is a single terminal).

Theorem: If L is a CFL, then $L - \{\epsilon\}$ has a CFG in CNF.

Review:

1. Eliminate ϵ -productions.
2. Eliminate unit productions.
3. Eliminate variables that derive no terminal string.
4. Eliminate variables not reached from the start symbol.

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